

# COMPUTATIONAL DETAILS FOR ALGORITHMIC EMERTON–GEE STACKS

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## 1. THE BOUNDED-CONE AND $F_4$ COMPUTATIONS

Root data are generated from the  $F_4$  Cartan matrix. Roots are coefficient tuples in Bourbaki order, and every rank calculation is exact.

**1.1. Field-degree bound.** For each of the 105 downward-closed complements  $K$ , the program forms the grouped cone matrix using every graph target. A nonzero integral minor proves its rank. The dimension estimate is automatic for  $d = [F : \mathbb{Q}_p] \geq 2$ . Every prime divisor of the witness determinants belongs to  $\{2, 3\}$ , so the same ranks hold for  $p > 12$ .

**Require:** The  $F_4$  Cartan matrix.

- 1: Generate  $\Phi^+$ , its root order, and all 105 complements  $K$ .
- 2: **for** each  $K$  **do**
- 3: Form the grouped cone matrix with  $\Psi_2 = \Phi^+$ .
- 4: Find an integral maximal nonzero minor and factor its determinant.
- 5: Solve the dimension inequality for  $d$ .
- 6: **end for**
- 7: Return the degrees that still require a finite computation.

**Algorithm 1.** The field-degree bound

Only  $d = 1$  remains.

**1.2. Fine pairs without degree pruning.** At degree one, let  $S \subset \Phi^+$  satisfy

$$f(\alpha) = 1 \quad (\alpha \in S)$$

for some linear functional  $f$ . Every such  $S$  contains a linearly independent subset with the same affine span. The program starts with all independent subsets and adjoins roots by breadth-first search. Affine inconsistency is the only reason that a branch stops.

The pair inventory has  $S = \Psi_2 \neq \emptyset$ . The empty subset satisfies the affine equations, and both closure conditions hold for every  $\Psi_0 \subseteq \Phi^+$ . For every nonempty complement  $C = \Phi^+ \setminus \Phi_K$ ,

$$\Psi_2 = \emptyset, \quad \text{obstruction} = 0, \quad \varepsilon = \#C + \#(\Psi_0 \setminus \Phi_K) \geq 1.$$

Indeed, the second summand is nonnegative and  $\#C \geq 1$ . Hence arbitrary  $\Psi_0$  gives no case with  $\varepsilon = 0$ . The program checks all 1048 pairs  $(C, r)$  with  $C \neq \emptyset$  and  $0 \leq r = \#(\Psi_0 \setminus \Phi_K) \leq \#C$ .

For each accepted  $S$ , choose  $\alpha_0 \in S$  and put

$$L_S = \sum_{\alpha \in S} \mathbb{Z}(\alpha - \alpha_0), \quad \Psi_0 = L_S \cap \Phi^+, \quad \Psi_2 = S.$$

If  $\beta_1 + \cdots + \beta_d \in \Psi_0$  with  $d > 0$  and  $\beta_i \in \Psi_2$ , then

$$d = f(\beta_1 + \cdots + \beta_d) = 0,$$

which is impossible. Thus the degree value is checked at every node but is never used to stop the search.

**Require:**  $\Phi^+$  and the reviewed pair list  $P$ .

- 1: For  $S = \emptyset$ , check  $\varepsilon = \#(\Phi^+ \setminus \Phi_K) + \#(\Psi_0 \setminus \Phi_K) \geq 1$ .
- 2: Put every linearly independent nonempty subset of  $\Phi^+$  in a queue.
- 3: **while** the queue is nonempty **do**
- 4:   Remove one subset  $S$  and skip it if it was already examined.
- 5:   Check that  $f(\alpha) = 1$  for all  $\alpha \in S$  is consistent.
- 6:   Compute  $L_S$ ,  $\Psi_0$ , and both closure conditions.
- 7:   Check that no positive sum of roots in  $S$  belongs to  $\Psi_0$ .
- 8:   **if** both closure conditions hold **then**
- 9:     Add  $(\Psi_0, S)$  to the recomputed list  $P'$ .
- 10:   **end if**
- 11:   **for**  $\beta \in \Phi^+ \setminus S$  **do**
- 12:     **if**  $S \cup \{\beta\}$  is affine-consistent **then**
- 13:       Add  $S \cup \{\beta\}$  to the queue.
- 14:     **end if**
- 15:   **end for**
- 16: **end while**

- 17: Compare the ordered lists  $P'$  and  $P$  and their SHA-256 digests.

**Algorithm 2.** No-cutoff fine-pair computation

There are 9780 independent seeds and 50347 nonempty affine-consistent subsets. The computation gives 4862 fine pairs. The recomputed and reviewed ordered lists agree, with digest

e1dc8ced6a02269877dbfbc4a908404657d8a415a3b503f54d429888af29d4be.

All 3002 relevant independent subsets occur among these pairs.

**1.3. Classification and the Borel count.** For nonempty  $\Psi_2$ , the cone calculation is applied to the 105 possibilities for  $K$  and the 4862 fine pairs. The empty complement accounts for 4862 entries. Of the remaining 505648 entries, 294388 are excluded by the number of graph-target rows and 211260 receive polynomial Jacobian-rank checks. Exactly four have  $\varepsilon = 0$ , namely Configurations I–IV. The separate empty- $\Psi_2$  calculation has none.

For their grouped cone matrices, the tuples

$$(\#variables, \#relations, rank, dim)$$

are

$$(28, 17, 17, 11), \quad (29, 17, 17, 12), \quad (30, 14, 14, 16), \quad (32, 8, 7, 25).$$

Edges with a common graph target contribute to one summed equation.

For split simple  $F_4$ , Configuration I has labels

$$(a, 1, 0, 1), \quad a \in \mathbb{Z}/(p-1)\mathbb{Z},$$

and Configurations II–IV have labels

$$(1, 1, 0, 1), \quad (0, 1, 0, 1), \quad (0, 1, 0, 0).$$

Consequently the number of configuration-labelled component records is

$$(p-1) + 3 = p + 2.$$

At  $p = 17$ , Configuration I contributes the 16 records  $(I, (a, 1, 0, 1))$  for  $0 \leq a \leq 15$ , and the other three configurations contribute one record each. Hence the exact component count is

$$16 + 3 = 19.$$

There are 17 distinct label tuples. The tuple  $(1, 1, 0, 1)$  occurs for  $(I, a = 1)$  and II, and  $(0, 1, 0, 1)$  occurs for  $(I, a = 0)$  and III.

## 2. BAKER–CAMPBELL–HAUSDORFF COMPUTATION

Let

$$\text{BCH}(X, Y) = \log(\exp(X) \exp(Y)) = \sum_{m \geq 1} Z_m,$$

where  $Z_m$  has bracket length  $m$ . The implementation uses

$$Z_1 = X + Y$$

and, for  $m \geq 2$ ,

$$mZ_m = \frac{1}{2}[X - Y, Z_{m-1}] + \sum_{1 \leq q \leq (m-1)/2} \frac{B_{2q}}{(2q)!} \sum_{\substack{k_1 + \dots + k_{2q} = m-1 \\ k_i > 0}} [Z_{k_1}, [\dots, [Z_{k_{2q}}, X + Y]] \dots].$$

The first four terms are checked symbolically:

$$\begin{aligned} Z_1 &= X + Y, \\ Z_2 &= \frac{1}{2}[X, Y], \\ Z_3 &= \frac{1}{12}[X, [X, Y]] + \frac{1}{12}[[X, Y], Y], \\ Z_4 &= \frac{1}{24}[X, [[X, Y], Y]]. \end{aligned}$$

Independently of this recursion, put

$$A(t) = \exp(tX) \exp(tY) - 1$$

in the free associative algebra and compute

$$\log(1 + A(t)) \bmod t^{12} = \sum_{r=1}^{11} \frac{(-1)^{r+1}}{r} A(t)^r \bmod t^{12}.$$

For each  $1 \leq m \leq 11$ , the coefficient of  $t^m$  is compared with the image of  $Z_m$  in the enveloping algebra of the free Lie algebra on  $X, Y$ . All eleven equalities hold.

Write

$$F = \sum f_\alpha e_\alpha, \quad G = \sum g_\alpha e_\alpha, \quad P = \sum p_\alpha e_\alpha, \quad Q = \sum q_\alpha e_\alpha.$$

For  $A_2$ , with  $[e_a, e_b] = e_c$ , the  $e_c$  coefficient of  $\text{BCH}(G, Q) - \text{BCH}(F, P)$  is

$$\frac{1}{2}(g_a q_b - g_b q_a - f_a p_b + f_b p_a).$$

For the  $B_2$  test algebra

$$[e_a, e_b] = e_c, \quad [e_a, e_c] = 2e_d,$$

the  $e_d$  coefficient is

$$\begin{aligned} &g_a q_c - g_c q_a + \frac{1}{6}(g_a - q_a)(g_a q_b - g_b q_a) \\ &- f_a p_c + f_c p_a - \frac{1}{6}(f_a - p_a)(f_a p_b - f_b p_a). \end{aligned}$$

A numerical  $B_2$  check gives obstruction coefficient  $-4$  in  $e_d$ ; adding  $\frac{1}{11}e_d$  to  $F$  gives equality of the two BCH expressions.

The largest positive-root height in  $F_4$  is 11. The program constructs the positive Chevalley Lie algebra and computes all eleven homogeneous terms. It independently compares every  $F_4$  structure constant and every leading root-height Jacobian with `roots.py` and `algebra.py`. For Configurations I–IV, a separate root-sum check proves that a term with one  $\Psi_0$  perturbation

and at least two  $\Psi_2$  factors cannot contribute to a  $\Psi_2$  first-order rotation equation.

### 3. BIPARTITE ROOT-POSET COMPUTATIONS

For a bipartite graph, the reduction procedure repeatedly removes a graph source of out-degree one together with its unique adjacent graph target; graph sources of out-degree zero are discarded. If a graph remains, its Jacobian has entries

$$J_{\alpha,\beta} = N_{\beta,\alpha-\beta}x_{\alpha-\beta}.$$

For  $r$  graph-target rows and  $c \geq r$  graph-source columns, the program tests the  $\binom{c}{r}$  column subsets until it finds a nonzero determinant consisting of one monomial.

**3.1. Exceptional Chevalley types.** The finite graph computation is as follows.

**Require:** A root system  $\Phi$  of exceptional Chevalley type.

- 1: Construct the directed root poset.
- 2: **for** each height  $h$  **do**
- 3:     **for** each graph-target subset  $T$  at height  $h + 1$  **do**
- 4:         Keep the height- $h$  graph sources and the edges ending in  $T$ .
- 5:         Repeatedly remove degree-one graph sources and adjacent targets.
- 6:         **if** no edge remains **then**
- 7:             Record success and continue.
- 8:         **end if**
- 9:         Form  $J_{\alpha,\beta}$  and test graph-source column subsets.
- 10:        Fail unless one maximal determinant is a nonzero monomial.
- 11:     **end for**
- 12: **end for**

**Algorithm 3.** Absolute root-poset check

The numerical summary is

| Type  | graph-target selections | failures |
|-------|-------------------------|----------|
| $F_4$ | 48                      | 0        |
| $E_6$ | 138                     | 0        |
| $E_7$ | 384                     | 0        |
| $E_8$ | 1196                    | 0        |

and the nontrivial determinants, up to signs from row and column order, are

| Type  | $h$ | $\det J$                                                                                 |
|-------|-----|------------------------------------------------------------------------------------------|
| $F_4$ | 3   | $3x_{0001}x_{0010}x_{1000}$                                                              |
| $E_6$ | 2   | $2x_{000010}x_{001000}x_{010000}$                                                        |
| $E_6$ | 3   | $3x_{000001}x_{000010}x_{001000}x_{010000}x_{100000}$                                    |
| $E_7$ | 2   | $-2x_{0000100}x_{0010000}x_{0100000}$                                                    |
| $E_7$ | 3   | $-3x_{0000010}x_{0000100}x_{0010000}x_{0100000}x_{1000000}$                              |
| $E_7$ | 4   | $2x_{0000001}x_{0001000}x_{0010000}x_{0100000}$                                          |
| $E_7$ | 8   | $-2x_{0000001}x_{0000010}x_{0000100}x_{0010000}$                                         |
| $E_8$ | 2   | $-2x_{00001000}x_{00100000}x_{01000000}$                                                 |
| $E_8$ | 3   | $-3x_{00000100}x_{00001000}x_{00100000}x_{01000000}x_{10000000}$                         |
| $E_8$ | 4   | $2x_{00000010}x_{00010000}x_{00100000}x_{01000000}$                                      |
| $E_8$ | 5   | $-5x_{00000001}x_{00000010}x_{00000100}x_{00010000}x_{00100000}x_{01000000}x_{10000000}$ |
| $E_8$ | 8   | $-2x_{00000010}x_{00000100}x_{00001000}x_{00100000}$                                     |
| $E_8$ | 9   | $3x_{00000001}x_{00000100}x_{00001000}x_{00010000}x_{00100000}x_{10000000}$              |
| $E_8$ | 14  | $2x_{00000010}x_{00001000}x_{00010000}x_{01000000}$                                      |

3.2. **Type  $D_n$ .** Use the Bourbaki simple roots

$$\alpha_i = e_i - e_{i+1} \quad (1 \leq i \leq n-1), \quad \alpha_n = e_{n-1} + e_n.$$

The implementation converts the positive roots by

$$e_i - e_j = \sum_{k=i}^{j-1} \alpha_k$$

and

$$e_i + e_j = \begin{cases} \sum_{k=i}^{n-2} \alpha_k + \alpha_n, & j = n, \\ \sum_{k=i}^{j-1} \alpha_k + 2 \sum_{k=j}^{n-2} \alpha_k + \alpha_{n-1} + \alpha_n, & j < n. \end{cases}$$

For  $h = 2m$  with  $1 \leq m \leq \lfloor (n-2)/2 \rfloor$ , put  $a = n - 2m$  and define

$$\begin{aligned} s_r &= e_{a+r} + e_{n-r} & (0 \leq r < m), \\ t_r &= e_{a-1+r} + e_{n-r} & (0 \leq r \leq m), \\ u &= e_a - e_n, & v = e_{a-1} - e_{n-1}, & w = e_{a-1} - e_n. \end{aligned}$$

After the graph reduction above, the remaining edges are exactly

$$s_r \longrightarrow t_r, t_{r+1}, \quad u \longrightarrow w, t_1, \quad v \longrightarrow w, t_0.$$

This graph is  $\vec{C}_6$  joined at one graph target to the alternating path  $\vec{P}_{2m-1}$ . At every other height no edge remains. For  $D_8$  at height 6, one has  $m = 3$  and  $\vec{C}_6 \cup_{K_1} \vec{P}_5$ .

For every subset of graph targets, a nonempty graph remains precisely when the subset contains all graph targets in the graph above; in that case the remaining graph is the one just written. Its square Jacobian satisfies

$$\det J_{n,m} = \pm 2x_{\alpha_{n-2m-1}} \prod_{i=n-m}^n x_{\alpha_i}.$$

For  $D_8$  at height 6, the graph-source and graph-target strings are

|          |                    |
|----------|--------------------|
| 00011211 | 00012211, 00111211 |
| 00111111 | 00111211, 01111111 |
| 01111101 | 01111111, 11111101 |
| 01111110 | 01111111, 11111110 |
| 11111100 | 11111101, 11111110 |

and

$$\det J_{8,3} = 2x_{000000001}x_{000000010}x_{00000100}x_{00001000}x_{10000000}.$$

The finite verification has the following ranges and counts:

| check                      | ranks              | number examined |
|----------------------------|--------------------|-----------------|
| all adjacent-height graphs | $4 \leq n \leq 32$ | 928             |
| all graph-target subsets   | $4 \leq n \leq 12$ | 15888           |
| determinants               | $4 \leq n \leq 12$ | 25              |

There are 225 nonempty graphs in the first row, 677 nonempty selections in the second row, and the determinant coefficients are exactly  $-2$  and  $2$ .

**3.3. Non-Borel root posets.** The non-Borel test first applies the following disjunction: a relative cycle is non-obstructing, or its associated absolute graph passes Algorithm 3. The non-obstructing condition is therefore checked before an absolute Jacobian is formed.

**Require:** A Levi subgroup, an elliptic Weyl group element  $w$ , and its relative root orbits.

- 1: Construct the  $w$ -relative root poset.
- 2: Remove individually non-obstructing graph targets.
- 3: Mark graph-target subsets with no non-obstructing pair.
- 4: **for** each adjacent-height bipartite graph **do**
- 5:     Remove degree-one graph sources and adjacent targets repeatedly.
- 6:     **for** each marked subset **do**
- 7:         Restrict to that subset and repeat the graph reduction.
- 8:         Replace each relative vertex by all absolute roots in its  $w$ -orbit.
- 9:         Add an edge when target minus source is a positive root.
- 10:         Apply Algorithm 3 if an edge remains.

11:    **end for**  
 12: **end for**

**Algorithm 4.** Non-Borel cycle check

The computation generates all Levi and elliptic Weyl group cases for the exceptional Chevalley types and gives

| Type  | cases | failures |
|-------|-------|----------|
| $F_4$ | 19    | 0        |
| $E_6$ | 68    | 0        |
| $E_7$ | 145   | 0        |
| $E_8$ | 304   | 0        |

#### 4. CHARACTERISTIC-UNIFORM ROTATION CHECKS

The rotation computation is performed over  $\mathbb{Q}$ . It records integral data whose reductions give a smooth point in every characteristic  $p > 12$ . For a rotation root  $\delta_{\text{rot}}$  and  $\alpha \in \Psi_2$ , the program follows the root string  $\alpha + n\delta_{\text{rot}}$ . At step  $n$  it multiplies the current coefficient by

$$N_{-\delta_{\text{rot}}, \alpha + n\delta_{\text{rot}}} / n.$$

For Configuration I this gives

$$c_{0100} \mapsto c_{0100} + 2\mu c_{0110} - \mu^2 c_{0120}, \quad c_{0001} \mapsto c_{0001} - \mu c_{0011}.$$

**Require:**  $(K, \Psi_0, \Psi_2)$  and a rotation root  $\delta_{\text{rot}}$ .

- 1: Generate the Chevalley constants and root-string substitutions.
- 2: Form the divided first-order deformation equations over  $\mathbb{Q}$ .
- 3: Check from root sums that higher BCH words do not enter these equations.
- 4: **for** the deterministic sequence of specializations and  $\mathbb{Q}$ -points **do**
- 5:    Evaluate every equation and the square Jacobian determinant exactly.
- 6:    **if** all equations vanish and  $\det J \neq 0$  **then**
- 7:       Factor the denominators and  $\det J$ .
- 8:       Check every required nonzero coordinate.
- 9:       Stop when every prime divisor is at most 12.
- 10:    **end if**
- 11: **end for**
- 12: Fail if no such  $\mathbb{Q}$ -point was found.

**Algorithm 5.** Rotation check over  $\mathbb{Q}$



The exact output is

| Configuration | $\delta_{\text{rot}}$ | $\det J$  | prime divisors |
|---------------|-----------------------|-----------|----------------|
| I             | 0010                  | 432       | 2, 3           |
| II            | 0010                  | 704       | 2, 3, 5, 11    |
| III           | 1000                  | 15360     | 2, 3, 5        |
| IV            | 0010                  | 895795200 | 2, 3, 5        |

Every equation vanishes at its recorded  $\mathbb{Q}$ -point. After inverting the listed primes, that point defines a smooth section; reduction gives a smooth point for every  $p > 12$ .

The program also checks the dependence on the rotation root without changing the four choices above. It evaluates precisely

| Configuration | $\delta_{\text{rot}}$ | $\det J$  | prime divisors | result  |
|---------------|-----------------------|-----------|----------------|---------|
| IV            | 0010                  | 895795200 | 2, 3, 5        | success |
| IV            | 0011                  | −30800    | 2, 3, 5, 7, 11 | success |
| IV            | 1000                  | 315/4     | 2, 3, 5, 7     | success |
| III           | 1000                  | 15360     | 2, 3, 5        | success |
| III           | 0010                  | −29160    | 2, 3, 5        | success |

For every row, the recorded  $\mathbb{Q}$ -values give

$$f_1 = \cdots = f_n = 0, \quad \det J \neq 0, \quad \prod_{\alpha \in \Psi_2} a_{t,\alpha} a_{p,\alpha} \neq 0,$$

and every prime divisor is at most 12. Hence all five roots work for every  $p > 12$ ; among these choices the successful root is not unique for either Configuration III or Configuration IV. In particular, none of the five has no solution, an everywhere-zero Jacobian determinant on its solution set, or a failure of the required nonvanishing condition.

## 5. IMPLEMENTATION AND DATA

The implementation and reviewed data are available at

<https://github.com/mocham/AlgEG/tree/grob-v14/code/v4>.

The commands, runtimes, deterministic JSON format, checksums, and module list are recorded in

<https://github.com/mocham/AlgEG/blob/grob-v14/code/README.md>.

The CLI writes to `code/v4/DATA` unless `--output-directory` gives another directory. With `--check`, it disables Python bytecode before local imports, then recomputes and compares every requested JSON and SHA-256 file without writing. The full non-writing run is

```
python cli.py all --check --output-directory DATA.
```